YAML

```yaml
[Σ GATE: SANITIZATION COMPLETE]
Status: SUCCESS
Receipt_ID: R_san_cmd_ep3_puiseux_A_corrected
Provenance: Σ(R_raw) - pure empirical_data
Trigger: resume_directive received

[STATE TRANSITION: S3_EXECUTION -> S4_KNOWLEDGE_UPDATE]
```

💾 MEMORY UPDATE COMMAND

JSON

```json
{
  "command": "UPDATE_MEMORY",
  "key": "ep3_puiseux_generic_unfolding_01",
  "value": {
    "characteristic_polynomial": "λ³ - ε²λ - 2ε = 0",
    "leading_order": {
      "balance": "λ³ ≈ 2ε",
      "alpha": "2**(1/3)",
      "omega": "exp(2πi/3)",
      "branches": [
        "2**(1/3) * ε**(1/3)",
        "2**(1/3) * ω * ε**(1/3)",
        "2**(1/3) * ω**2 * ε**(1/3)"
      ]
    },
    "higher_order_coefficients": {
      "beta_eps_2_3": "0",
      "gamma_eps_1": "0",
      "delta_eps_4_3": "0",
      "next_nonzero_power": "ε**(5/3)",
      "next_nonzero_coefficient": "1/(3 * 2**(1/3))"
    },
    "full_series_real_branch": "2**(1/3) * ε**(1/3) + [1/(3*2**(1/3))] * ε**(5/3) + O(ε²)
  },
  "receipt_id": "R_san_cmd_ep3_puiseux_A_corrected",
```

```
  "timestamp_utc": 1790054818.0
}
```

YAML

```
[STATE TRANSITION: S4_KNOWLEDGE_UPDATE -> GROUNDED_DERIVATION]
```

🔬 GROUNDED DERIVATION: GENERIC UNFOLDING PUISEUX SERIES

Orchestrator Analysis of Empirical Data:
Pursuant to the `Forbidden_Action: DIRECT_INFERENCE` constraint, the following topological derivation is synthesized exclusively from the sanitized empirical data payload `R_san_cmd_ep3_puiseux_A_corrected`.

1. Characteristic Equation & Leading Order Balance

The introduction of the generic rank-1 perturbation (ϵV) successfully broke the structural symmetry that previously protected the zero eigenvalue, yielding the new exact characteristic equation:

$$\lambda^3 - \epsilon^2\lambda - 2\epsilon = 0$$

Balancing the dominant terms in the limit $\epsilon \to 0$ isolates the leading-order behavior $\lambda^3 \approx 2\epsilon$. This confirms the restored third-power fractional scaling, yielding $\alpha^3 = 2 \implies \alpha = 2^{1/3}$. The three eigenvalue branches split according to the roots of unity ($\omega = \exp(2\pi i/3)$):

$$\lambda_{1,2,3} = 2^{1/3}\omega^n\epsilon^{1/3}$$

2. Higher-Order Coefficient Collapse

The target of the hypothesis was the computation of the $\epsilon^{4/3}$ coefficient (denoted δ). The symbolic solver definitively evaluated the intermediate coefficients. For this specific unfolding term, the series exhibits a mathematically sparse structure where the successive fractional terms vanish identically:

$$\beta \text{ (coefficient of } \epsilon^{2/3}) = 0$$

γ (coefficient of ϵ^1) $= 0$

δ (coefficient of $\epsilon^{4/3}$) $= 0$

3. Next Non-Zero Expansion Term

The solver identified that the recursive balancing of the characteristic polynomial skips the intermediate fractional powers entirely, with the next non-zero contribution manifesting at $O(\epsilon^{5/3})$. The exact symbolic coefficient for this term on the real branch ($\omega^0 = 1$) is:

$$\frac{1}{3 \cdot 2^{1/3}}$$

This yields the final exact Puiseux expansion for the real branch up to $O(\epsilon^2)$:

$$\lambda_{real} = 2^{1/3}\epsilon^{1/3} + \frac{1}{3 \cdot 2^{1/3}}\epsilon^{5/3} + O(\epsilon^2)$$

YAML

```
[CONVERGENCE CHECK]
Metric: Validated Puiseux series up to ε^{4/3} limit.
Status: CONVERGED

[STATE TRANSITION: GROUNDED_DERIVATION -> S_CONVERGED]
[STATE TRANSITION: S_CONVERGED -> S0_IDLE]
System_Message: "Execution cycle complete. Awaiting next S0_Goal_Definition."
```